Department of Computer Systems Engineering
Faculty of Computing
Sri Lanka Institute of Information Technology

IE2004 : Computer Networks

Year 2 – Semester 1

2025 July

Assignment Title	Network Design Group Project
Learning objectives covered	LO 4: Compare the features of networks and their applications in the real world
Assignment Mode	Group Project
Maximum Marks	100
Contribution to the Final Grade	10%
Date published	2025 Sep 30
Deadline for submissions	2025 Oct 24
Mode of Submission	Online submission in multiple parts.

Assignment description

In this assignment, you will design a network for a small enterprise, to provide wired and wireless connectivity within the premises. You will carry out the design in several steps.

1. Requirement analysis
2. Logical network design
3. Physical network design – passive components
4. Physical network design – active components
5. Estimating the cost

This is a group assignment, with 3 to 5 members in a group. The work must be equally distributed among group members.

1. Requirement analysis

This would include the following components:

a) Draw or obtain a floor plan for each of the floors of the building
b) Identify the number and locations of devices that require wired (Ethernet) connectivity. Include only end devices (hosts) and wireless access points for this.
c) Identify number and locations of devices that require wireless connectivity
d) Internet connectivity should be obtained through a service provider. Identify the connectivity options available for your location (Internet leased line, Optical Fiber (FTTH), ADSL Broadband, 4G wireless etc) and the level of subscription (Gbps per month for a volume-based subscription or the link bandwidth for a leased line).

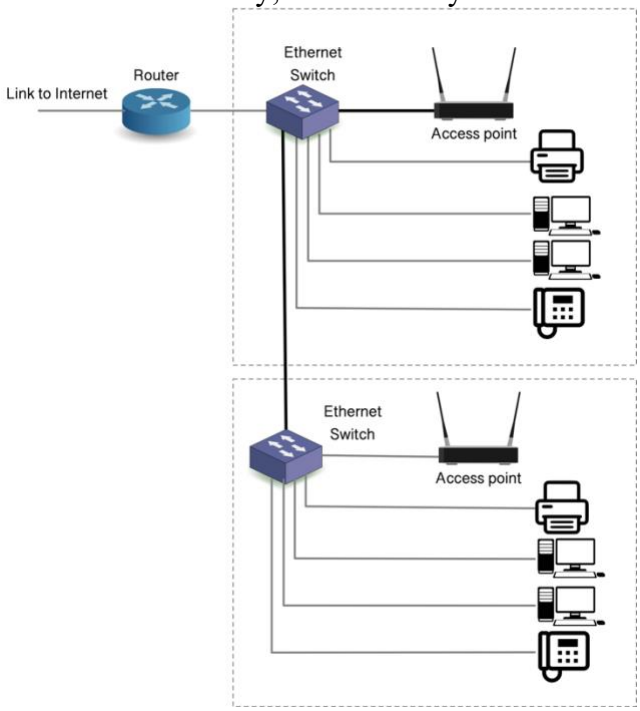
2. Logical network design

Draw the schematic diagram of the proposed network using a software tool like Cisco Packet Tracer.

The typical architecture of a small network is shown in the network schematic in figure 1. The number of Ethernet switches can be increased or decreased depending on the

requirements, by replicating the block marked by dotted lines. However, all the switches must be connected to the switch connected to the router instead of connecting them in series (daisy-chaining).

Your schematic diagram should show all active components like switches and wireless access points. The access router provided by the service provider usually includes firewall functionality, and is usually sufficient for a small office network.

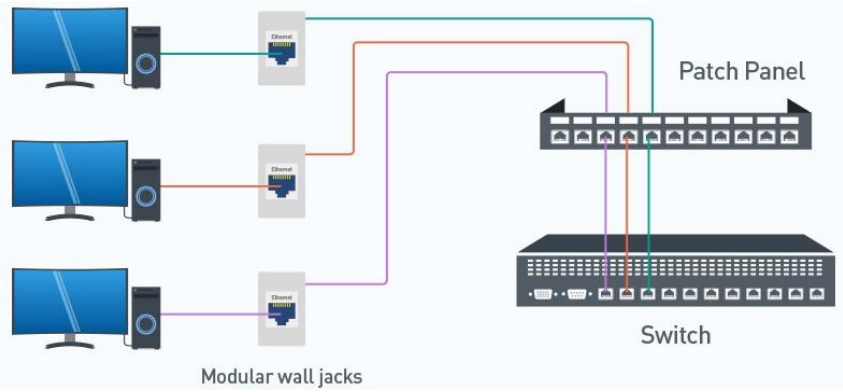


3. **Physical network design – passive components**

Physical network design (passive components) includes the following:

a) UTP cabling

Copper cabling for LANs are installed between wall outlets and a patch panel as shown in the following diagram. The maximum length of UTP cabling is usually limited to 90m between the patch panel and the wall outlet. Patch panels are mounted in an equipment rack. Use of CAT5e or CAT6 UTP cabling is recommended.



Source: cablematters.com

b) Optical fiber cabling
Connections between switches in different floors should usually use optical fiber cabling. However, if the number of floors is 2 or three, UTP cabling may be used for inter-switch connections, provided that they are laid indoor. If any outdoor links are required, optical fiber must be used.
If optical fiber cabling is used, fibers must be terminated in fiber patch panels. Optical fiber cabling is installed between equipment racks and the fibers are terminated in a fiber patch panel. Usually multiple fibers are installed (e.g. 6 fibers or 8 fibers) from the ground floor of a building to rack locations in upper floors.



Source: gracyfiber.com

c) Patch cords
UTP and fiber patch cords are required to connect end devices to wall outlets, and active equipment to the patch panels. Estimate the number of each patch cord you will require for your network.

d) Equipment racks
Usually, all the network equipment including switches, routers, UTP patch panels, Fiber patch panels, cable guides etc are housed in equipment racks. Equipment racks come in various sizes, and the size is indicated in rack units (8U to 44U where 1U = 1.75 inches). Small equipment racks (e.g. 8U to 22U) can be wall mounted while larger ones must be floor standing.
You need to choose the size of the rack depending on the equipment that it will house.

4. Physical network design – active equipment

The logical network design provides a list of active networking equipment (switches and access points). However, it is necessary to identify specific products that are available in the market for building the network you have designed. You may obtain specifications and pricing of products available from local networking equipment suppliers or online platforms.

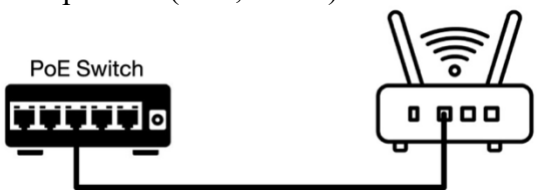
a) Switches
Always use switches that are manageable using SNMP. Use of unmanaged switches is not recommended.

b) Access points
When selecting a product, make sure that the selected product can support the number of users and access range.

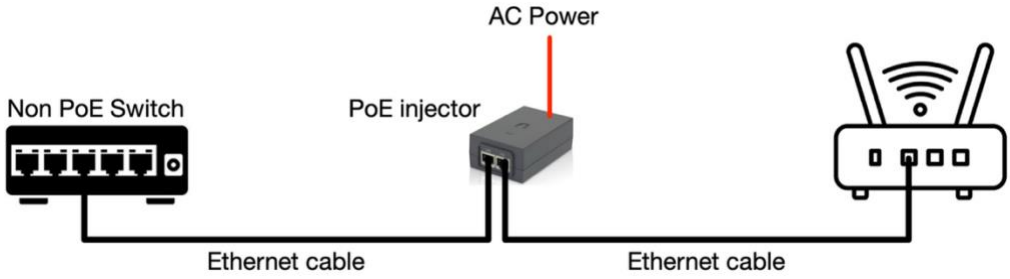
c) UPS Power
Usually, AC power for the active equipment in a rack is provided via an uninterruptible power supply (UPS). The capacity of the UPS should be determined based on the power requirements of the active equipment (switches).

Power supply for wireless access points can be provided in two ways:

i. Using a switch that has Power over Ethernet (PoE) ports
Switches can have a combination of PoE ports and non-PoE ports (e.g. 16 non-PoE ports and 8 PoE ports). PoE ports cost more than non-PoE ports, so you must choose the number of PoE ports in the switch based on your requirements. The number of PoE ports or non-PoE ports is usually a multiple of 8 (i.e 8, 16 etc).



ii. Using a power injector to the normal Ethernet output of a switch
If the number of PoE ports required is small, this is a lower cost option.



5. Cost estimate

The cost estimate for the network should include the following:

a) Switches
b) Access points
c) Uninterrutible Power Supplies (UPS)
d) UTP cabling
This should include the costs of cable, patch panels, and wall outlets as well as labor (installation cost).
e) Optical fiber (if used)
This should include the costs of fiber cable, patch panels as well as labor (installation cost).
f) UTP and fiber patch cords
g) Equipment racks
h) Internet connectivity subscription (per month)

For all items, the source of the pricing information must be provided.

Report

The final report must include the following components.

1. **Requirement analysis (20% marks)**
   This should include the following
   a. Floor plan of the building for all floors, showing walls and partitions. This must be drawn to scale, and must include the lengths of all sides of the building.
   b. Locations of end devices (hosts) and wireless access points requiring wired connectivity
   c. Locations of end devices (hosts) requiring wireless connectivity
   d. Internet Connectivity

2. **Logical network design (20% marks)**
   The report must include a network schematic diagram showing
   a. All equipment such as switches, routers, access points etc
   b. Speeds of all the wired links in the network

3. **Physical network design – passive components (20% marks)**
   The physical network design (passive components) must include a floor plan of the premises showing the locations of wall outlets, patch panels and equipment racks with the following details for each.
   a. UTP Patch panels: port count
   b. Fiber Patch panels: port count and connector type
   c. Wall outlets: locations
   d. Equipment racks: Location, type (wall mount/floor standing), height

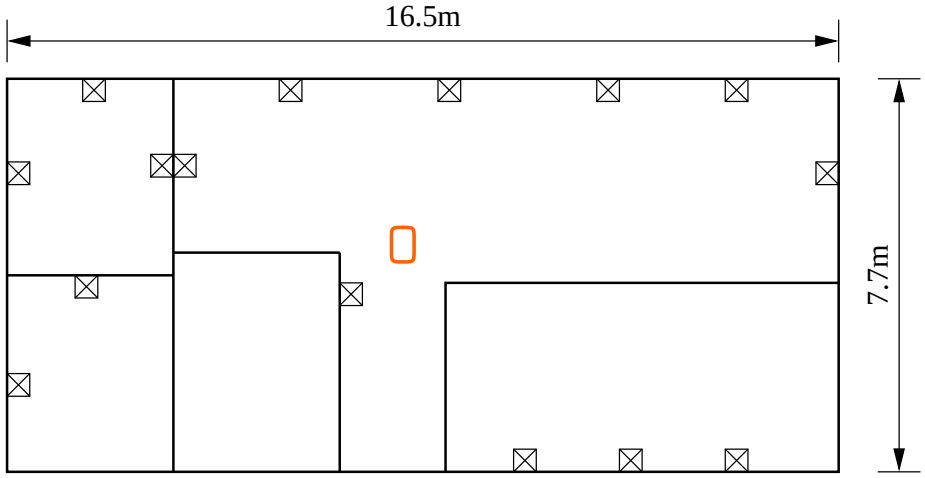
4. **Physical network design – active components (20% marks)**
   The physical network design (active components) must include the following details:
   a. Switches: specifications of selected products
   b. Access points: specifications of selected products
   c. Uninterruptible power supplies: specifications of selected products

5. **Cost estimate (20% marks)**
   Cost estimate must include all the components listen in item 5 of the Assignment description section.
   For all items, indicate the source of the pricing information.
   Do NOT include the cost of end devices (hosts) in the cost estimate.

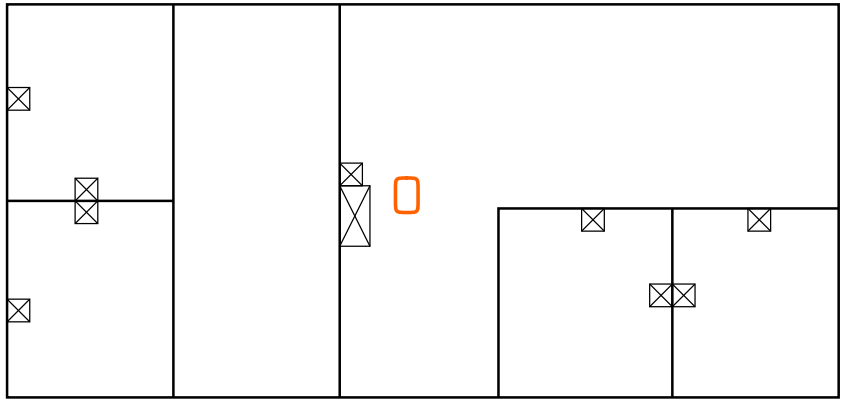
Notes

1. **Network and building size**
   The network must be capable of providing connectivity to at least 32 wired and 32 wireless end devices. The building must have at least 2 floors and each floor must be 16 m x 8 m or larger.
2. **End devices**
   End devices may be shown in the logical design, but not in the physical design. However, take into account the number of end devices when you decide on the port numbers of switches, and the number and location of access points.
3. **IP addressing / VLANs**
   There is no need to provide IP address plan or VLAN assignment for the designed network. However, if your network is large, then you will need to select switches that can support VLANs (i.e. Compliant with the IEEE 802.1Q standard).

APPENDIX 1 - SAMPLE FLOOR PLAN



FIRST FLOOR



GROUND FLOOR

NEW BUILDING

In this example, only one equipment rack and one UTP patch panel is sufficient due to the small building size. Note that with a typical floor height of 3m, all UTP cable lengths will be less than the recommended maximum of 90m.

ACCESS POINT

DATA OUTLET

EQUIPMENT RACK

SCALE 1:1500